PubChem Cross-Validation for ModelSEED Compounds

Full-Dataset Validation and Correction Results

ModelSEED Database Cleanup

April 9, 2026

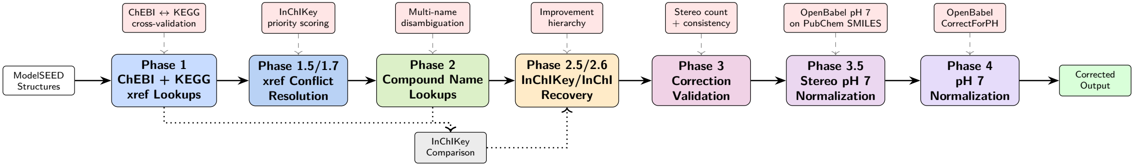
Dataset Overview

Metric	Count
Total compounds	36,844
With structures	30,323
With names	45,171
Validated	**30,321**

Goal: Validate whether compound structures in ModelSEED correctly correspond to their PubChem records, and apply evidence-based corrections.

Approach: Multi-phase pipeline that queries PubChem using cross-references (ChEBI/KEGG), compound names, and direct structure lookups (InChIKey/InChI), then classifies discrepancies and applies validated corrections.

Scope: 30,321 compounds with both a searchable identifier and a stored InChIKey.



Each phase queries PubChem for the compound's InChIKey, which is compared block-by-block against the stored InChIKey to classify the result. Phases 2.5/2.6 recover unresolved compounds via direct structure lookups. Phase 3.5 normalizes accepted STEREO_DIFF corrections to pH 7, preventing PubChem's neutral protonation from overwriting curated ionization states. Phase 4 normalizes remaining compounds' protonation to pH 7. All results are cached in SQLite for resumability.

InChIKey structure (three hyphen-separated blocks):

$\underbrace{\texttt{BSYNRYMUTXBXSQ}}_{\texttt{Connectivity (14)}} - \underbrace{\texttt{UHFFFAOYSA}}_{\texttt{Stereo (10)}} - \underbrace{\texttt{N}}_{\texttt{P}}$

Block 1: connectivity (atoms, bonds). Block 2: stereo (E/Z, R/S). Block 3: protonation (N = neutral).

Classification by block comparison:

Category	Blk 1	Blk 2	Blk 3
MATCH	=	=	=
PROTONATION_DIFF	=	=	$\neq$
STEREO_DIFF	=	$\neq$	--
MISMATCH	$\neq$	--	--

Interpretation:

- MATCH: Identical molecule.
- PROT. DIFF: Same molecule, different protonation state. Represents the same compound at different pH.
- STEREO_DIFF: Same connectivity, different stereochemistry. May indicate missing stereocenters.
- MISMATCH: Different molecular graph — incorrect mapping or tautomeric difference.

Hierarchical comparison (Block 1 first, then 2, then 3) identifies the most significant differences first. Only STEREO_DIFF is eligible for automated correction.

Methodology: PubChem Lookup Strategies (1/2)

Phase 1: Cross-reference lookups

- Query PubChem by ChEBI and KEGG registry IDs
- Both identifiers queried when available — conflicting CIDs → `XREF_CONFLICT`
- Highest confidence (curated mappings)

Phase 1.5/1.7: Conflict resolution

- Resolve ChEBI≠KEGG conflicts by comparing candidate InChIKeys to stored InChIKey
- Priority scoring: exact match > block 1+2 > block 1
- Remaining conflicts: fetch full CID properties

Phase 2: Name lookups

- For compounds without xrefs or failed lookups
- Up to 3 names per compound (scored by quality)
- Multi-name disambiguation via InChIKey matching

Phase 2.5: InChIKey recovery

- For NOT_FOUND, MISMATCH, or STEREO_DIFF after Phases 1–2
- Query PubChem with the **stored InChIKey**
- 97.0% match rate — indicates initial discrepancies were due to incorrect mappings, not incorrect structures

Methodology: PubChem Lookup Strategies (2/2)

Phase 2.6: InChI recovery

- For still-unresolved compounds after Phase 2.5
- Direct InChI lookup in PubChem

Phase 3: Correction validation

- Fetch full CID properties for STEREO_DIFF and MISMATCH
- Apply validation gate (stereo counting, consistency)
- Cache validated corrections in SQLite

Phase 3.5: Stereo pH 7 normalization

- `CorrectForPH(7.0)` on accepted STEREO_DIFF SMILES
- Prevents PubChem neutral protonation from overwriting curated ionization states

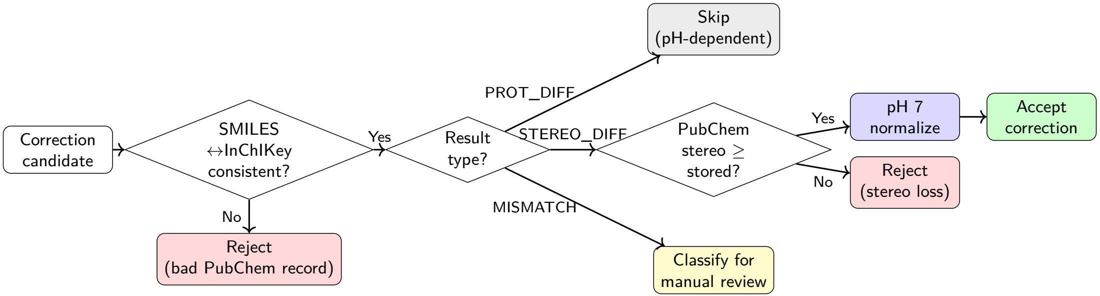
Phase 4: pH 7 normalization

- OpenBabel `CorrectForPH(7.0)` on remaining compounds
- Only apply if protonation-only change (no connectivity change)
- Rejects free phosphate over-deprotonation (pK_{a4}>7)

Improvement hierarchy (Phase 2.5):

- NOT_FOUND → any result
- MISMATCH → MATCH, PROT_DIFF, or STEREO_DIFF
- STEREO_DIFF → MATCH or PROT_DIFF

Methodology: Correction Validation Gate

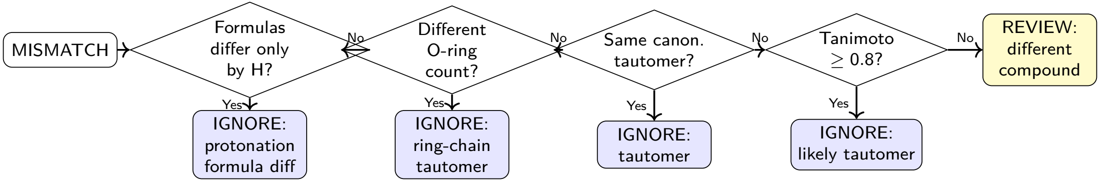


STEREO_DIFF: Accepted only if PubChem defines $\geq$ stereocenters than the stored structure *and* shared stereocenters are not inverted (InChI stereo layer comparison). Accepted corrections are then normalized to pH 7 (OpenBabel) to preserve the curated ionization state.

MISMATCH sub-classification (all for manual review):
Formula gate $\rightarrow$ ring-chain tautomer check $\rightarrow$ canonical tautomer check (RDKit) $\rightarrow$ Tanimoto similarity ($\geq$ 0.8).

Methodology: MISMATCH Sub-Classification

Different connectivity (Block 1) = different molecular graph — never auto-corrected. Each MISMATCH is classified to prioritize manual review:



IGNORE: the stored ModelSEED structure represents a different tautomeric form than PubChem's canonical representation. No correction needed.
REVIEW: same molecular formula but low structural similarity — may indicate an incorrect cross-reference mapping to a positional or constitutional isomer.

Validation Results

Category	Count	%
MATCH	19,984	65.9
PROTONATION_DIFF	4,365	14.4
XREF_CONFLICT	2,919	9.6
MISMATCH	1,331	4.4
NOT_FOUND	832	2.7
STEREO_DIFF	683	2.3
AMBIGUOUS	152	0.5
Other	55	0.2
Total	**30,321**	**100**

Interpretation:

- **80.3%** (MATCH + PROT_DIFF) share the same connectivity and stereochemistry with PubChem
- **9.6%** XREF_CONFLICT — ChEBI and KEGG cross-references point to different PubChem CIDs
- **4.4%** MISMATCH — different molecular connectivity; flagged for manual review
- **2.3%** STEREO_DIFF — same connectivity, different stereochemistry assignments
- **2.7%** NOT_FOUND — no matching PubChem record identified
- **0.5%** AMBIGUOUS — compound names resolve to multiple distinct PubChem CIDs

Lookup Strategy Breakdown

Strategy	Count
KEGG xref	9,717
Name lookup	8,981
ChEBI xref	6,245
Xref conflict	2,919
InChIKey recovery	2,228
Ambiguous	152
Xref resolved	79
Total	**30,321**

Agreement rate by strategy (MATCH + PROT_DIFF):

Strategy	Agree	Rate
InChIKey recovery	2,161 / 2,228	97.0%
ChEBI xref	5,924 / 6,245	94.9%
Name lookup	7,411 / 8,149	90.9%
KEGG xref	8,775 / 9,717	90.3%
Xref resolved	78 / 79	98.7%

InChIKey recovery (Phase 2.5) yields the highest match rate at 97%. Among initial lookup strategies, ChEBI xrefs have the highest agreement rate (94.9%). KEGG xrefs produce the most protonation differences.

Phase 2.5 re-queries PubChem using the stored InChIKey for compounds that were NOT_FOUND, MISMATCH, or STEREO_DIFF after Phases 1–2. This recovers cases where the stored structure is correct but the cross-reference or name mapping was wrong.

Result	Count
MATCH	2,161 (97.0%)
STEREO_DIFF	51 (2.3%)
MISMATCH	2 (0.1%)
Unknown	14 (0.6%)
Total	**2,228**

Interpretation:

2,228 compounds that were previously unresolved or mismatched were recovered, of which 97% returned exact InChIKey matches.

This indicates that for these compounds, the initial discrepancies were due to incorrect cross-reference mappings or name ambiguity rather than incorrect stored structures.

Without Phase 2.5: these compounds would remain classified as NOT_FOUND or MISMATCH.

Automated corrections applied only to STEREO_DIFF compounds. MISMATCH compounds classified for manual review (never auto-corrected).

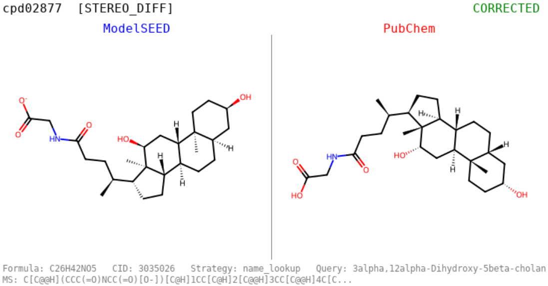
Category	Total	Corrected	Rejected	Action
STEREO_DIFF	683	188	495	Auto: PubChem $\geq$ stored stereo + pH 7
PH7 normalization	30,321	13	1	Phase 4: OpenBabel pH 7
MISMATCH	1,331	0	1,331	Classified for manual review
PROTONATION_DIFF	4,365	0	0	Skipped (pH-dependent)
XREF_CONFLICT	2,919	0	0	Unresolved (ChEBI $\neq$ KEGG)
AMBIGUOUS	152	0	0	Skipped (names disagree)
NOT_FOUND	832	0	0	No PubChem reference

Total corrections applied: 201 compounds (188 stereochemistry + 13 pH 7 protonation).

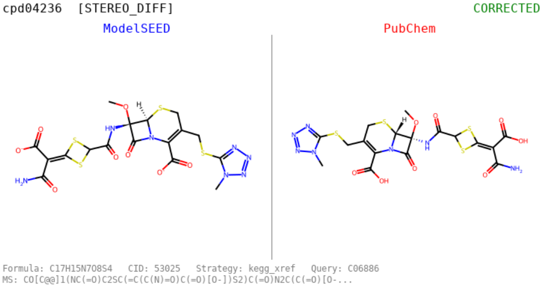
STEREO_DIFF corrections (188): each is pH 7-normalized (Phase 3.5) before application so PubChem's neutral protonation does not overwrite the curated ionization state. **Rejections** (495): fewer stereocenters or inversions.

PH7 (Phase 4): deprotonates phosphate/sulfate groups; free (non-ester) phosphates excluded (p$K_{a4} > 7$).

cpd02877 — Glycodeoxycholate



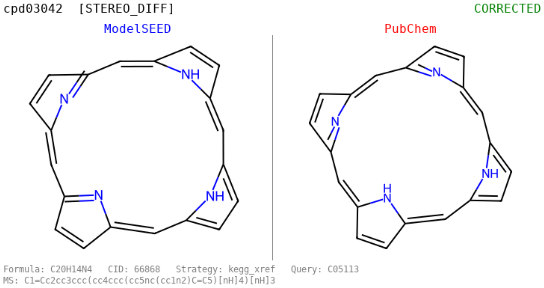
cpd04236 — Cefotetan



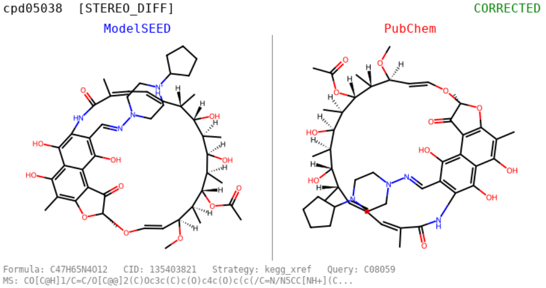
cpd02877 (Glycodeoxycholate, name lookup). PubChem defines all stereocenters for the bile acid scaffold. Accepted — no existing stereocenters inverted.

cpd04236 (Cefotetan, KEGG xref). PubChem defines additional stereocenters on the cephalosporin ring. Accepted — adds missing stereochemistry.

cpd03042 — Porphyrin



cpd05038 — Rifapentine

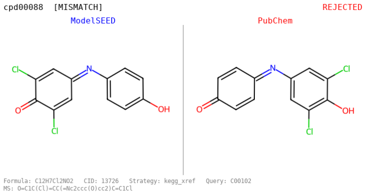


cpd03042 (Porphyrin, KEGG xref). PubChem replaces the `UHFFFAOYSA` achiral placeholder with a defined stereo block. Accepted.

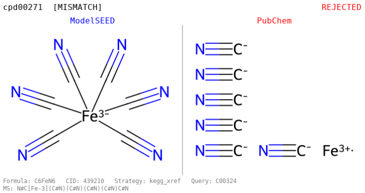
cpd05038 (Rifapentine, KEGG xref). Macrolide antibiotic; PubChem record defines additional stereocenters. Accepted.

MISMATCH Examples: Tautomers and Wrong Mappings

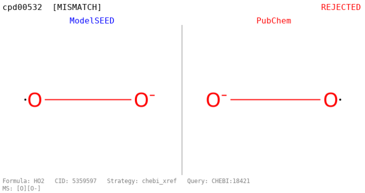
cpd00088 — DCIP



cpd00271 — Ferricyanide



cpd00532 — Superoxide



cpd00088 (Dichloroindophenol, KEGG:C00102).
KEGG xref maps to a structurally different compound in PubChem.
REVIEW:different_compound.

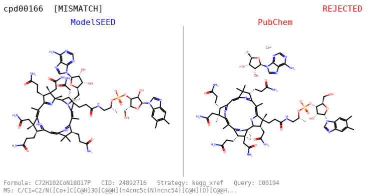
cpd00271 (Ferricyanide, KEGG:C00324).
PubChem record has different connectivity. Rejected — requires manual review.

cpd00532 (Superoxide, ChEBI:18421).
Molecular formula differs between stored and PubChem structures.
Rejected.

All MISMATCH compounds are classified for manual review. None are auto-corrected.

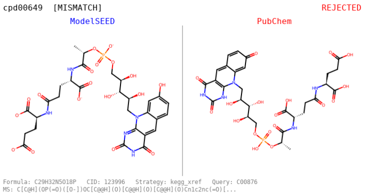
MISMATCH Examples: More Cases

cpd00166 — B12



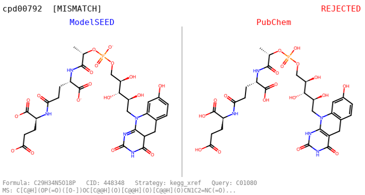
cpd00166 (Adenosylcobamide, KEGG:C00194).
Cobalt-containing cofactor; stored and PubChem structures differ. Rejected — requires manual review.

cpd00649 — Coenzyme F420



cpd00649 (Coenzyme F420, KEGG:C00876).
Methanogenic cofactor; PubChem record has different connectivity. Rejected.

cpd00792 — Reduced F420



cpd00792 (Reduced F420, KEGG:C01080).
Reduced form of F420; same connectivity discrepancy as cpd00649. Rejected.

Phase 4 normalizes compounds to their pH 7 microspecies using OpenBabel's `CorrectForPH(7.0)`. Only protonation-only changes are accepted (same connectivity). Free (non-ester) phosphate over-deprotonation is rejected ($\mathrm{p}K_{\mathrm{a}4} \approx 9.4 > 7$).

Compound	Name	Old Chg	New Chg	Change
cpd00002	ATP	-3	-4	Deprotonated phosphate
cpd00008	ADP	-2	-3	Deprotonated phosphate
cpd00014	UDP	-2	-3	Deprotonated phosphate
cpd00031	GDP	-2	-3	Deprotonated phosphate
cpd00038	GTP	-3	-4	Deprotonated phosphate
cpd00081	Sulfite	-1	-2	Deprotonated sulfite
Rejected:				
cpd00012	Diphosphate (PPi)	-3	—	Free phosphate $\mathrm{p}K_{\mathrm{a}4}{=}9.4$

13 corrections are single-proton deprotonations of ester-linked phosphate or sulfate groups ($\mathrm{p}K_{\mathrm{a}} \approx 6.5$). Free PPi (cpd00012) rejected: $\mathrm{p}K_{\mathrm{a}4}{=}9.4$, proton remains at pH 7. Connectivity (InChIKey Block 1) unchanged in every case.

Key findings:

- **80.3%** (MATCH + PROT_DIFF) share the same connectivity and stereochemistry with PubChem
- **Phase 2.5 recovered 2,228 compounds** (97% match rate) — initial discrepancies traced to incorrect mappings
- **188 stereo corrections applied** — PubChem adds missing stereochemistry
- **13 pH 7 normalizations** — deprotonation of phosphate/sulfate (free PPi excluded, $pK_{a4}=9.4$)
- **1,331 MISMATCH classified** — for manual review
- **2,919 xref conflicts** remain unresolved (ChEBI $\neq$ KEGG)

Next steps:

1. Resolve remaining XREF_CONFLICT compounds (2,919)
2. Manual review of MISMATCH compounds (1,331)
3. Manual review of AMBIGUOUS compounds (152)
4. Investigate NOT_FOUND compounds (832) for missing xrefs
5. Run full Phase 4 pH 7 normalization on complete dataset

Appendix

Detailed Category Analysis & Technical Details

Data Quality by Compound ID Range

ID Range	N	Match	Prot.	Mis.	Stereo	N/F	Amb.	Xref
0–9,999	9,463	49.1%	21.9%	3.9%	2.6%	0.0%	0.0%	22.1%
10k–19,999	4,166	65.4%	15.7%	4.6%	2.3%	0.2%	0.0%	11.5%
20k–29,999	5,773	76.2%	11.7%	3.3%	2.7%	0.6%	1.1%	4.2%
30k–39,999	6,814	80.2%	9.1%	6.2%	1.5%	0.4%	1.1%	1.5%
40k+	4,105	66.8%	8.5%	3.7%	2.1%	18.4%	0.4%	0.0%

- Core metabolites (0–9,999) have high xref conflict rate (22.1%) and protonation diffs (21.9%) — heavily cross-referenced
- Compounds >40,000 have 18.4% NOT_FOUND — recently added with sparse external references
- Match rate improves in mid-ranges (20k–40k) — 76–80%
- MISMATCH rate peaks in 30k–40k range (6.2%)

By lookup strategy:

Strategy	Mismatches
KEGG xref	600 (45.1%)
Name lookup	523 (39.3%)
ChEBI xref	206 (15.5%)
InChIKey lookup	2 (0.2%)
Total	**1,331**

KEGG xrefs account for 45.1% of all mismatches, followed by name lookups (39.3%). ChEBI xrefs produce the fewest mismatches (15.5%). InChIKey recovery yields only 2 mismatches out of 2,228 lookups.

Stereo diffs by strategy:

Strategy	Count
KEGG xref	315 (46.1%)
Name lookup	215 (31.5%)
ChEBI xref	102 (14.9%)
InChIKey lookup	51 (7.5%)
Total	**683**

KEGG xrefs are the largest source of stereo differences (46.1%). 188 of 683 STEREO_DIFF compounds passed validation and were corrected (27.5%).

Protonation Difference Analysis

By lookup strategy:

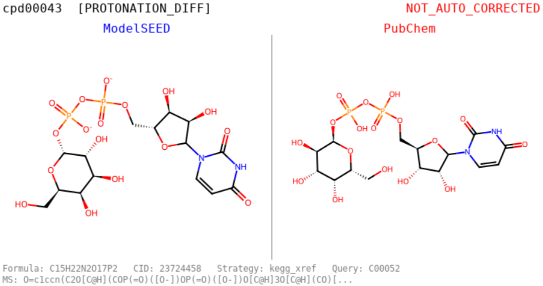
Strategy	Count	Rate
KEGG xref	2,880	29.6%
Name lookup	1,324	16.3%
ChEBI xref	144	2.3%
Xref resolved	17	21.5%
Total	**4,365**	

Interpretation:

- KEGG produces 20× more protonation diffs than ChEBI
- ModelSEED stores pH 7 charge states; PubChem stores neutral forms
- **Not auto-corrected**: the difference reflects different pH conventions, not an error
- Phase 4 pH 7 normalization addresses compounds where ModelSEED's stored protonation was inconsistent with pH 7

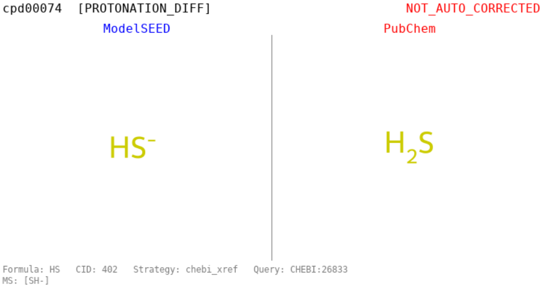
Example	Compound	ModelSEED	PubChem
cpd00002	ATP	ATP^{3-} (partially protonated)	ATP^{4-} (fully deprotonated)
cpd00012	Diphosphate	PP_i^{3-} (one P-OH)	PP_i^{4-} (fully ionized)

cpd00043 — UDP-Galactose



ModelSEED stores the pH 7 protonation state. PubChem stores a different protonation state. Skipped — same compound at different pH.

cpd00074 — Selenocysteine



Different protonation state: ModelSEED stores the pH 7 form, PubChem stores the neutral form. Skipped.

Not auto-corrected: the protonation difference reflects the intended pH 7 representation in ModelSEED vs. the neutral form in PubChem.

2,919 compounds have conflicting ChEBI and KEGG cross-references pointing to different PubChem CIDs. Only 79 resolved by Phase 1.5/1.7.

Resolution strategy:

- Compare candidate InChIKeys to stored InChIKey
- Priority: exact match > blocks 1+2 > block 1
- Source preference: ChEBI > KEGG
- Phase 1.7: fetch full CID properties for remaining

Why so many unresolved?

- Core metabolites (0–9k) most affected (22.1%)
- Many xrefs map to different protonation forms
- Neither candidate matches stored InChIKey exactly

Example (cpd00059):

ChEBI → CID 54679076 vs KEGG C00072 → CID 54670067

Both CIDs correspond to PubChem records for the same compound but with different charge states or stereoisomers. Requires manual review to determine which is correct.

Impact: 2,919 compounds cannot be validated until conflicts are resolved. Concentrated in the 0–10k ID range where compounds have the most cross-references.

Safety Features and Validation Checks

Quality Gates

- Formula cross-check for MISMATCH
- Stereo center counting (RDKit) for STEREO_DIFF
- InChI stereo layer comparison (no inversions)
- SMILES↔InChIKey consistency check
- ChEBI/KEGG xref cross-validation
- Multi-name ambiguity detection
- pH 7 connectivity check (no bond changes)

Data Preservation

- Never overwrites original structures file
- Corrections log with ISO 8601 timestamps
- Formula/Charge recomputed from corrected InChI
- Atomic writes (temp file + rename)

Cache and Resumability

- SQLite cache (CIDs, InChIKeys, SMILES)
- `--clear-cache` for fresh runs
- Rate-limited PubChem API (5 req/s)
- Thread-safe conflict tracking

Output files

- `pubchem_validation_report.tsv`
- `pubchem_mismatches.tsv`
- `pubchem_stereo_diffs.tsv`
- `pubchem_protonation_diffs.tsv`
- `pubchem_corrections_log.tsv`
- `pubchem_rejected_corrections.tsv`
- `pubchem_protonation_corrections.tsv`
- `pubchem_review_different_compounds.tsv`